

Dormitory Management System

Entity Relationship Diagram (ERD)

Database Schema Version 1.0 | Updated: September 2026

USERS	
PK id	bigint
name	varchar(255)
email	varchar(255)
phone	varchar(20)
password	varchar(255)
is_active	boolean
last_login_at	timestamp

ROOMS	
PK id	bigint
room_number	varchar(10)
floor	tinyint
capacity	tinyint
monthly_rate	decimal(10,2)
is_airconditioned	boolean
meter_number	varchar(30)
status	varchar(20)

TENANTS	
PK id	bigint
FK user_id	bigint

CONTRACTS	
PK id	bigint
FK room_id	bigint
start_date	date
end_date	date
due_day	tinyint
deposit_required	decimal(10,2)
deposit_collected	decimal(10,2)
status	varchar(20)
is_active	boolean
FK created_by	bigint

CONTRACT_TENANTS	
PK id	bigint
FK contract_id	bigint
FK tenant_id	bigint
joined_on	date
left_on	date

DEPOSIT_ENTRIES	
PK id	bigint
FK contract_id	bigint
type	varchar(20)
amount	decimal(10,2)
date	date
reason	varchar(255)
FK recorded_by	bigint

INVOICES	
PK id	bigint
invoice_number	varchar(30)
FK contract_id	bigint
billing_month	varchar(7)
due_date	date
tenant_count	tinyint
rent_amount	decimal(10,2)
electricity_amount	decimal(10,2)
carry_over_balance	decimal(10,2)
credit_balance	decimal(10,2)
total_amount	decimal(10,2)
amount_paid	decimal(10,2)

TENANT_PAYMENTS	
PK id	bigint
FK invoice_id	bigint
FK tenant_id	bigint
share_amount	decimal(10,2)
amount_paid	decimal(10,2)
status	varchar(20)
method	varchar(30)
paid_at	timestamp

		status varchar(20) FK created_by bigint	
PAYMENTS PK id bigint FK invoice_id bigint FK tenant_payment_id bigint amount decimal(10,2) method varchar(30) received_at date reference varchar(100) FK recorded_by_user bigint	METER_READINGS PK id bigint FK room_id bigint reading_month varchar(7) previous_kwh decimal(10,2) current_kwh decimal(10,2) kwh_used decimal(10,2) FK recorded_by bigint	SERVICES PK id bigint code varchar(20) name varchar(100) weight_limit_kg decimal(5,2) is_active boolean	SERVICE_PRICES PK id bigint FK service_id bigint customer_type enum price decimal(8,2)
LAUNDRY_ORDERS PK id bigint FK tenant_id bigint order_date timestamp pickup_date date status varchar(20) notes text	ORDER_ITEMS PK id bigint FK laundry_order_id bigint FK service_id bigint quantity int price decimal(8,2) subtotal decimal(10,2)	SETTINGS PK id bigint key varchar(100) value text effective_from date FK set_by bigint	AUDIT_LOGS PK id bigint FK user_id bigint action varchar(50) model_type varchar(100) model_id bigint description text

Entity Relationships

Users → Tenants: One-to-One (1:1) Each user can be one tenant, each tenant has one user account	Rooms → Contracts: One-to-Many (1:N) One room can have multiple contracts over time
Contracts → Tenants: Many-to-Many (M:N) Multiple tenants can share one contract via contract_tenants	Contracts → Deposit Entries: One-to-Many (1:N) One contract can have multiple deposit transactions
Contracts → Invoices: One-to-Many (1:N) One contract generates monthly invoices	Invoices → Tenant Payments: One-to-Many (1:N) One invoice split into payments per tenant
Invoices → Payments: One-to-Many (1:N) One invoice can have multiple payment transactions	Rooms → Meter Readings: One-to-Many (1:N) One room has monthly electricity readings

Tenants → Laundry Orders: One-to-Many (1:N)
One tenant can place multiple laundry orders

Services → Service Prices: One-to-Many (1:N)
One service has different prices per customer type

Laundry Orders → Order Items: One-to-Many (1:N)
One order contains multiple service items

Users → Multiple Entities: One-to-Many (1:N)
Users create/record contracts, invoices, payments, readings

Legend

- PK** Primary Key - Unique identifier
- FK** Foreign Key - Reference to another table
- Regular Field - Data attribute